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EXPERIMENT TITLE: U-TUBE HYDROSTATIC PRESSURE

Objectives :

1. Determining the density of a fluid based on the principle of pressure balance in a U-tube

EXPERIMENT SUMMARY

INTRODUCTION

Fluid is one of the substances that we often encounter in our daily lives, both in the form of liquids and gases. One of the properties of static fluid is that it can exert pressure in all directions caused by the weight of the fluid itself (Halliday et al, 2014). This concept is then used in various purposes in daily life, ranging from distributing water to homes through water towers, dams designed to be able to withstand water pressure at a certain depth, then infusions in hospitals that use pressure differences to regulate the rate of blood flow and also tools to measure pressure such as U-tube manometers that are widely used in piping systems in industry.

Hydrostatic pressure itself is the pressure exerted by a fluid that is in a state of idleness due to the influence of gravitational force. The magnitude of the hydrostatic pressure itself depends on the density of the fluid, the acceleration of gravity and the height of the fluid. Mathematically, hydrostatic pressure can be expressed through the following equation

$$P = P_0 + \rho g h \tag{1}$$

where P is the pressure at the depth of h, $P0$ is the pressure on the open surface of the fluid (atmospheric pressure), ρ is the density of the fluid, g is the acceleration of gravity, and h is the depth of a fluid. This equation shows that the pressure exerted by the fluid will increase as the depth and density of the fluid increases (Young & Freedman, 2020). Density is one of the physical properties of a substance that expresses the amount of mass contained in each unit of volume. Each substance has a different density depending on the type of material it is constituting, so density can be used as one of the differentiators of a substance from other substances (Mikrajuddin, 2016)

One simple tool that utilizes the principle of hydrostatic pressure is the U-tube as an example shown in Figure 1. The working principle of the U-tube is based on the state of pressure equilibrium, i.e. if two fluids that do not mix with each other are in a state of rest in the tube, then the pressure at two points that have the same height will be of the same value. Thus, if point A and point B are on one horizontal plane, then a relationship applies

$$P_A = P_B \tag{2}$$

and the pressure at each point can be expressed as

$$P_A = P_0 + \rho_1 g h_1 \tag{3}$$

and

$$P_B = P_0 + \rho_2 g h_2 \tag{4}$$

With ρ_1 and ρ_2 being the density of fluid 1 and fluid 2, while h_1 and h_2 are the height of fluid 1 and fluid 2 to the reference point in the tube. Since both fluid surfaces are open to the atmosphere, the atmospheric pressure (P_0) on both sides eliminates each other so that a relationship is obtained

$$\rho_1 g h_1 = \rho_2 g h_2 \tag{5}$$

since the acceleration of gravity in both segments is considered the same, Equation (5) can be simplified to

$$\rho_1 h_1 = \rho_2 h_2 \tag{6}$$

Equation (6) is the basis for determining the density of a fluid using the principle of hydrostatic pressure of the U-tube (Serway & Jewett, 2018).

METHODOLOGY

A. Tools and Materials

The following are the tools and materials used in this practicum:

Table 1. Tools and materials for the long measuring tool

Tools	Quantity	Function
U Tube	2 pieces	Used to measure fluid height
Syringe	1 piece	Used to insert aqueducts into U-tubes
Pipette	2 pieces	Used to insert fluid into a U-tube
Tripod	1 piece	Used for thermometer support
Beaker Glass	3 pieces	Used as a container for fluids
Termometer	1 piece	Used to measure room temperature
Aquatics	Enough	Used as a comparator fluid
Isopropyl Miristat	Enough	Used as a density tested fluid
Baby Oil	Enough	Used as a density tested fluid

B. Working Steps

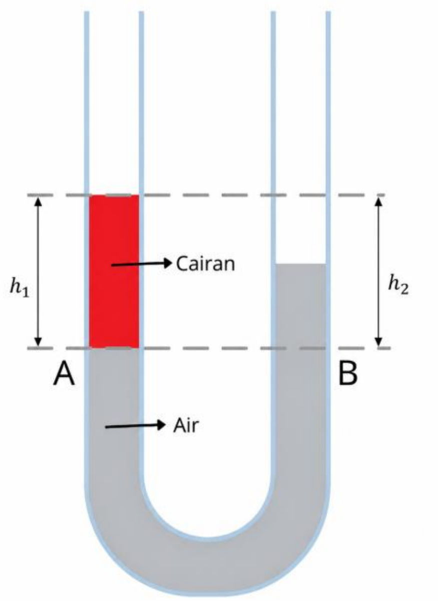

Figure 1. Schematic on U-tube hydrostatic pressure experiments

The working steps of measuring using calipers are as follows:

1. Prepare tools and materials.
2. Measure the temperature of the room using an alcohol thermometer.
3. Prepare aquaade, isopropyl myristat and *baby oil* and pour in the available container in sufficient quantities.
4. Put the aqueduct into a 10 mL U-tube using *a syringe* (not necessarily exact).
5. Put about 30 drops (not necessarily exact) of isopropyl myristate into a U-tube, then measure y1, y2, and y3.
6. Repeat the 5th step 5 times.
7. Discard all the fluid in the U-tube, clean it, then repeat steps 4 to 6 2 times.
8. Do steps 4 to 7 using *baby oil.*

D. Experiment Flowchart

DATA ANALYSIS AND DISCUSSION

A. Analysis of Measurement Data
The following is a table of data from the results of measurements made in the laboratory:

Table 1. The results of the measurement of fluid height in Isopropyl Miristat at temperature....°C

Repetition to-	y_1 (cm)	y_2 (cm)	y_3 (cm)
1			
2			

Table 2. The results of the measurement of fluid height in *Baby Oil* at temperature....°C

Repetition to-	y_1 (mm)	y_2 (cm)	y_3 (cm)
1			
2			

Approved by

Laboratory Assistant

B. Data Analysis of Calculation Results
(Find the value of the 2nd fluid density using Equation 6, and compare it to the theoretical value)
Example calculation:

The following is a table of the data obtained from the calculations:

Table 3. Results of Isopropyl Myrisat Density Calculation.

Repetition to-	h_1 (cm)	h_2 (cm)	$\rho_{IPM}(g/cm^3)$
1			
2			
Average			
Standard deviation			
Uncertainty			

Table 4. Results of Baby Oil Density Calculation.

Repetition to-	h_1 (cm)	h_2 (cm)	$\rho_{Baby\ Oil}(g/cm^3)$
1			
2			
Average			
Standard deviation			
Uncertainty			

C. Chart Analysis
(Make a linear regression graph and write down the equation)

D. Discussion

CONCLUSION

From this experiment it can be concluded:

1.

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Serway, R.A. & Jewett, J.W., 2018. *Physics for Scientists and Engineers with Modern Physics* (10th ed). Boston, MA: Cengage Learning.

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PRELIMINARY TASKS

1. Why does the liquid to be measured in density have to be hydrophobic?
2. What fluid should be put in a U-tube? Why?
3. If water and liquid have been put into the U-tube, what will happen when data collection is done with the addition of water instead of liquid?
4. What is the meniscus? How do you measure the height of a fluid that is experiencing the meniscus?
5. What happens to the hydrostatic pressure of a liquid if its depth increases? Explain